A Cloth Simulation Method Based on Differential Geometric Energy

Li Liu[1 2 3], Ruomei Wang[1 2 4], Zhuo Su[1 2], Fengwei Wang[1 2]
[1] *National Engineering Research Center of Digital Life, State-Province Joint Laboratory of Digital Home Interactive Applications, School of Information Science & Technology, Sun Yat-sen University, Guangzhou 510006, China*
[2] *Shenzhen Digital Home Key Technology Engineering Laboratory, Research Institute of Sun Yat-sen University in Shenzhen, Shenzhen 518057, China*
[3] *Faculty of Information Engineering and Automation, Kunming University of Science and Technology, Kunming 650500, China*
[4] *Email: isswrm@mail.sysu.edu.cn*

***Abstract*—According to the properties of bending and stretching in the cloth simulation, in order to achieve a superior cloth deformation effects, this paper proposes a new cloth simulation method based on the geometric deformation model that can preserve geometric features in the deformation better. Based on differential domain constraints condition, deformation energy function based on the different types of cloth materials is optimized. Nonlinear least squares deformation energy is the core of this differential framework. The cloth deformation is simulated interactively by increasing the related constraints and iteration methods. On the basis of design of geometric energy, transformation, and the corresponding nonlinear optimization numerical solution, the cloth deformation can be simulated more realistic. Especially, the geometric features about textile material effects can be simulated.**

***Keywords*-cloth simulation, mesh deformation, geometric features, geometric energy**

I. Introduction

Cloth simulation is an important research issue and has wide applications in computer graphics, geometric modeling, computer animation, video games and virtual clothing. Cloth simulation is a complex processes, it involves cloth modeling, numerical computation algorithm and physical properties of textile material. Around this area, some references have reported their research results. Up to now, the efficiency of simulation algorithm, computational cost, interaction complexity and deformation quality still are hot research problems in the world.

A. Related work

Cloth simulation still remains a challenge because most real-world clothes exhibit dynamic and complex wrinkles and folds. Many of methods with respect to this topic addressed, such as physically-based simulation, geometrically-based simulation and mesh deformation. In this section we review previous work briefly.

1) Physically-based cloth simulation: Breen et al. [1] developed a non-continuum particle model for cloth drape that explicitly represents the micro-mechanical structure of a fabric according to physical laws. In 1995, Provot [2] proposed a mass-spring system for clothing simulation. Baraff et al. [3] proposed the implicit integral cloth simulation method based on spring-particle system. Recently, Kang et al. [4] proposed a way to use a harmonic republican function to calculate the overall deformation effect and also use the finite element method (FFM) to get the local surface details on cloth. Popa et al. [5] proposed a method of using space-time deformation to simulate dynamic effects on clothes with continuous wrinkles and folds by utilizing 16 cameras. Later, Aguiar [6] and Wang [7] introduced a real-time machine learning method to obtain the wrinkles and folds of cloth, but these methods also limit their proper scope because it cannot produce wrinkles and folds without training data. In summary, the simulating objects in physically-based simulation are different due to their corresponding distinctive woven nature in the course of cloth simulation, creating the models requires finely tuned physical parameters and the expensive computation.

2) Geometrically-based cloth simulation: There have been wide development and application of geometric-based methods in cloth simulation owing to their high efficiency of numerical algorithms. Liu [8] and Chen [9] introduced a full geometric approach for developable objects and cloth simulation, however, these methods can only get the final shape of the cloth, but not its dynamic behavior. Goldenthal et al. [10] presented an elaborate numerical solver for cloth simulation based on a framework of constrained Lagrangian mechanics. Bridson et al. [11] proposed edge-based quantities to simulate cloth dynamics. But none of these methods considered cloth material properties. It is difficult to achieve the visual pleasing and realistic effects during the process of cloth simulation.

3) Mesh deformation: Keeping the local geometric details during the process of deformation is the main problem of mesh deformation algorithm. The mesh editing algorithms [12] based on differential theory are new methods developed in recent years. The geometric meaning of these algorithms

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is clear, simple, high efficient with recognition of morphing and other mesh editing functions. These methods [13] based on Laplacian coordinates are the typical representations. It can obtain mesh deformation results with a more clear and concise preservation to surface average curvature using optimized solution procedure. However, the Laplacian coordinates are limited in description of the other geometric features, e.g., the edge length preservation, straightness and volume preservation.

In order to control the deformation, a cloth simulation method based on differential geometric energy is presented in this paper. The cloth can be defined as the square original mesh (Fig. 1(a)), we can specify the fixing vertices A, B, C, D around the flat mesh and then lift the intermediate vertex as a controlling point S for direct mesh deformation based on the Laplacian coordinates, it can be only produced deformations effects like membrane deformation (Fig. 1(b)) lacking of the geometric details.

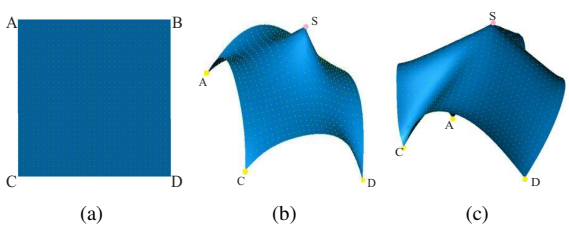

Figure 1. (a) Input plain meshes with 961 vertices. (b) The result of Laplacian coordinates with fixing the vertices (A, B, C, D) of four corners and dragging the central vertex (S) as additional vertex position constraint for deformation. (c) shows the result of our method for cloth deformation.

B. Framework of our method

Our method is highly motivated by the surface deformation method [15] based on differential coordinates and actually obtains the cloth deformation effects via the geometric energy. The local geometric features are preserved in our method can be visualized in Fig. 1(c). The framework of our cloth simulation method is presented in Fig. 2.

In order to preserve the local surface geometric details in the course of cloth deformation efficiently, according to the characteristics of cloth geometric deformation, the total geometric energy including the constraint factors of position, length, angle and gravity constraints are constructed. Then the geometric energy based on these terms is optimized in the numerical solution to improve the computational efficiently. Because our deformation method is a large nonlinear system, we set up an iterative procedure to obtain the dynamic effects of cloth simulation. Finally, the experiments show that our method can obtain expected simulation results.

The main contributions of our method are as follows.
(1) Construct the geometric deformation energy function for cloth simulation with respect to the cloth materials.
(2) Optimize the geometric energy for cloth simulation with a numerical solution in the linear least square sense.

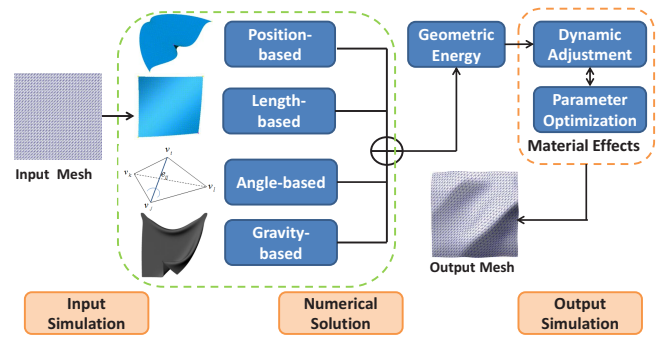

Figure 2. The framework of our method, including these major steps, geometric energy construction, numerical solution, dynamic adjustment and optimization procedure.

(3) Propose a dynamic adjustment scheme to simulate cloth deformation with cloth material effects.
(4) Demonstrate the applicability of our method to several applications in cloth simulation.

The rest of this paper is organized as follows. In the section 2, current differential geometric methods are summarized, the cloth deformation measure is proposed. In the section 3 an iterative optimization procedure and adjustment scheme based on the nonlinear formulation are introduced. In the section 4 several deformation results based on our cloth simulation method are given. Finally, the conclusion and future work are given in Section 5.

II. Geometric Energy in Cloth Deformation

The aim of mesh deformation is to preserve geometric details efficiently. To preserve the local geometric details of the cloth mesh during the deformation, we firstly construct geometric energy function based on differential domain theory, and then solve by assembling these energy terms into a unified optimal calculation framework. The details of our method are stated in the following.

At the beginning, the cloth is built as an initial triangular mesh $M = (V, E, F)$ (in Fig. 3(a)). Here, V is the vertex set, E is the edge set, F is the triangle strips set. Let $V = [v_1, v_2, ..., v_n], v_i \in R^3$ and $V' = [v_1', v_2', ..., v_n'], v_i' \in R^3$ denote the set of vertices of the initial mesh M and the target mesh M' respectively. Each vertex is composed of its X, Y and Z coordinates v_i=$[x_i, y_i, z_i]$ and v_i'=$[x_i', y_i', z_i']$.

A. Position-based energy

According to the requirement of cloth mesh deformation, the final target mesh M' after optimization should be similar to its edited mesh M. Because the interactive manipulation is adopted during a cloth simulation, if we know the position of vertices on the mesh surface, the error between the actual position of the control vertex and the target position designated by the user can be minimized by virtue of position constraint. The position constraint has two purposes, one is that some fixed vertices on mesh are used to keep its existing shape,

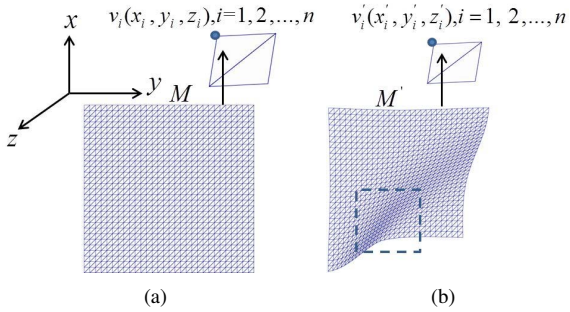

Figure 3. (a) The initial cloth mesh. (b) The target mesh is the deformation result of only vertex preservation, the marked regions show the initial mesh size are compressed after deformation.

and the other is moved in terms of user's requirements in each deformation. We can get the first approximation energy form of the objective function:

$$E_p = \sum_{i=0}^{n} \left\| v_i^{'} - v_i \right\|^2. \tag{1}$$

This item restrains that the current vertices v_i are approximated to the target mesh vertices $v_i^{'}$. The difference between current and target vertices can be obtained by the Euclidian distance measurement. However, only the elastic membrane effect can be produced without length-preservation and the initial cloth mesh is stretched, as shown in Fig. 3(b).

B. Length-based energy

Most real-world cloth surfaces are flexible due to the inextensibility of cloth. In order to achieve deformation effect like cloth, we encode differential quantities at edges at new deformed vertex positions instead of encoding local information at vertices. With a set of constraints (the transformation of edge E_e), it can be obtained by the following least square minimization problem

$$E_e = \sum_{e \in E} \left| \ell(e^{'}) - \ell(e) \right|^2, \tag{2}$$

where E is the edge set, $\ell(e)$ is the length of edge e on the initial mesh M, $\ell(e^{'})$ is the length of edge $e^{'}$ on the target mesh $M^{'}$. For an edge e, its length is determined by its two vertices v_i and v_j respectively. And the length of edge $e^{'}$ depends on the other two vertices $v_i^{'}$ and $v_j^{'}$. The numeric value of $\ell(e)$ and $\ell(e^{'})$ can be achieved by the following formula

$$\begin{cases} \ell(e) = \|v_i - v_j\| \\ \ell(e^{'}) = \left\| v_{\mathrm{i}}^{'} - v_{\mathrm{j}}^{'} \right\| . \end{cases}$$

C. Angle-based energy

In addition to the above two measures, another measure using to simulate the rich geometric details (e.g. folds and wrinkles) is taken into account

$$E_a = \sum_{e \in E} \left\| \theta(e^{'}) - \theta(e) \right\|^2. \tag{3}$$

Where $\theta(e^{'})$ and $\theta(e)$ are the magnitude related to the dihedral angle of two adjacent triangles on initial mesh $M^{'}$ and the target mesh M. Given an edge e_{ij} , let vertex v_k and v_l be the third vertex in the two triangles that share e_{ij} respectively (see Fig. 2). We can calculate the magnitude $\theta(e^{'})$ and $\theta(e)$ with $(\lambda v_i^{'} + (1-\lambda)v_j^{'}) - (\mu v_k^{'} + (1-\mu)v_l^{'})$ and $(\lambda v_i + (1-\lambda)v_j) - (\mu v_k + (1-\mu)v_l)$. The coefficients λ and μ are chosen to make $\theta(e^{'})$ and $\theta(e)$ parallel to normal of the edge e_{ij}. The deformation process gradually decreases the total interpolation error E_a until it becomes zero or the termination conditions of the iteration are met.

D. Gravity-based energy

The traditional processing methods based on geometry have no gravitation potential energy when manipulating surfaces. But it can be easier to extend our methods for global force constraints in this paper. Especially, E_g is used for the purpose of simulating gravity, an optional potential energy term can be added into the energy function, $\mathbf{g}$ is a vector of stress direction and the magnitude of gravity, E_g is defined as

$$E_g = < \mathbf{g}, v_i >, \tag{4}$$

where $V = \{ v_i \in R^3 | 1 < i < n \}$.

E. The deformation energy

Vertices coordinates of the meshes are obtained finally while the new vertices coordinates are calculated by minimizing the deformation energy function which defined as $V = \arg\min_V E(V)$, the final geometric formula is $E(V) = E_p + E_e + E_a + E_g$. All of the four terms are combined together to obtain the operation of cloth mesh deformation. And there are some important system control parameters, such as these weights α, β and δ which are greater than 0. The cloth mesh deformation procedure, starting from the initial M, eventually reaches the shape when all E_p, E_e, E_a and E_g are minimum. Then, we aimed at solving the following minimization problem, the total energy function is

$$V = \arg\min_V (\alpha E_p + \beta E_e + \delta E_a + E_g), \tag{5}$$

the goal of cloth simulation is to optimize this total geometric energy function.

III. Cloth simulation with material effects

We construct an iterative procedure of dynamic weights adjustment scheme for cloth simulation, we also use material property description into our cloth simulation to obtain different kinds of realistic cloth effects.

A. Numerical solution

Iterative schemes are generally used for non linear optimization. We iteratively linearize the objective function with respect to variables and changes in a non linear least squares problem to be the linear one. Given the initial configuration and constraints, implicit parameters and explicit vertex positions iteratively can also be calculated until they meet termination conditions. There are $3 \times n$ variables in Eq. 5. The corresponding coordinates of n vertices in vertex set V are (x_i, y_i, z_i). These variables are expressed by the column vector of $X = [x_1, y_1, z_1, x_2, y_2, z_2, ..., x_n, y_n, z_n]^T$. Hence, the three geometric energy terms are expressed as geometric constraints on X:

$$\begin{cases} \rho(v_i) = v_i^{'} - v_i = 0, & 1 \leq i \leq n, \\ \delta(e) = \ell(e^{'}) - \ell(e) = 0, & e \in E, \\ \sigma(v_i) = \theta(e^{'}) - \theta(e) = 0, & e \in E. \end{cases} \tag{6}$$

The left-hand sides of all the equations in (6) are the functions of X. These equations are highly nonlinear and overdetermined. And thus the goal of our approach is developed to iteratively linearize the equation (6) and then obtain the results solved by the least square method. It is noted that the matrix $A^T A$ is sparse, positive definite and symmetric. Therefore, it is possible to compute a Cholesky factorization of $A^T A$, which helps to solve (6) more efficiently.We can solve it by the least square sense, $A^T A X^{'} = A^T b$. Because the Hesse-Matrix of E_g is always zero, the potential energy contributes to the linear equations on the right side only.

B. Our algorithm and dynamic adjustment

We give a general algorithm framework of the cloth mesh deformation. The whole cloth simulation process based on the deformation energy with objective function is an iterative optimization process. The pseudo code of this scheme is given in the table of Algorithm 1.

Like any iterative algorithm, the three weights α, β and δ are not static and play an important role in our algorithm, they are updated dynamically at each iteration. Under some circumstances, designers may want to adjust cloth materials dynamically for intervening during the deformation procedure shown in Algorithm 2. To facilitate gauging the progress of the energy reduction, the changes of three terms including E_p, E_e and E_a should be maintained over the last m iterations (where $m = 10$ in our implementation).

Algorithm 1: Cloth Simulation Scheme

Input: M: initial mesh, m: iterations
Output: $M^{'}$: target mesh
Begin
If(Termination) **then** Exit
else do Algorithm 2 determine the three weights α, β, δ
for $n = 0$ to m do
$M \leftarrow$ variable of v_i coordinates on initial mesh
$M^{'} \leftarrow$ variable of $v_i^{'}$ coordinates on target mesh
$\ell(e) \leftarrow$ Euclidean distance of (x_i, y_i, z_i) and (x_j, y_j, z_j)
$\ell(e^{'}) \leftarrow$ Euclidean distance of $(x_i^{'}, y_i^{'}, z_i^{'})$ and $(x_j^{'}, y_j^{'}, z_j^{'})$
$\theta(e), \theta(e^{'}) \leftarrow$ construct the angle value
end for
for $n = 0$ to m do
$v_i^{'} - v_i$ linearize position-based energy
$\ell(e^{'}) - \ell(e)$ linearize edge-based energy
$\theta(e^{'}) - \theta(e)$ linearize angle-based energy
X=$(A^{\mathrm{T}}A)^{-1}A^{\mathrm{T}}b$ lest square numerical solution
$M^m = M^{m+1}$
end for
return
End

Algorithm 2: Dynamic Adjustment Scheme

Input: M: initial mesh, α: position-based weight,
β: edge-based weight, δ: angle-based weight
Output: $M^{'}$: target mesh
Begin
if $(r_p < minimum_rate_E_p)$ **then do** $\alpha \leftarrow 1.75\alpha$
if $(r_e < minimum_rate_E_e)$ **then do** $\beta \leftarrow 1.5\beta$
if $(r_a < minimum_rate_E_a)$ **then do** $\delta \leftarrow 1.25\delta$
initial $\alpha \leftarrow \frac{\alpha}{\alpha+\beta+\delta} \beta \leftarrow \frac{\beta}{\alpha+\beta+\delta} \delta \leftarrow \frac{\delta}{\alpha+\beta+\delta}$
do $\beta \leftarrow 1e-7$
if $(\alpha > \alpha^{\max})$ **then do** $\alpha \leftarrow \alpha^{\max}$
if $(\beta > \beta^{\max})$ **then do** $\beta \leftarrow \beta^{\max}$
if $(\delta > \delta^{\max})$ **then do** $\delta \leftarrow \delta^{\max}$
return 1
End

IV. Results and discussion

In this section, a prototype of our proposed cloth simulation algorithm will be implemented and compared with the related typical differential geometric method. The effectiveness of our approach have been demonstrated in experiments. All the tests were performed on a notebook with Intel i3-2100 3.10GHz CPU and 4GB RAM, Visual Studio 2008 and Graphite software [16].

Fig. 4 shows the comparison of our method to other approaches. Most of the previous deformation methods based on differential domain can only consider using average curvature preservation. It only gets the membrane-like effect of deformation result, see Fig. 4(a). The result in Fig. 4(b) based on space deformation is limited in representing the geometric features. As shown in Fig. 4(c), the wrinkles on the local surface can be produced. When we add the gravity-based energy, the realistic effect exhibit in Fig. 4(d).

Fig. 5 illustrates the effectiveness of three energy terms including vertex position, edge length and angle for cloth simulation. The rich deformation effects are achieved by adjusting the weights to balance the maintaining of curvature and the keeping of length and angle. Fig. 6 compares our algorithm result to other main processed algorithm in reference paper [15] and shows the improvement results. Fig. 7 illustrates deforming an initial flat piece of cloth mesh by using dynamic adjustments of coefficients α, β and δ. Fig. 7(a)-(c) are the results without the gravity energy. Fig. 7(d)-(f) show the effects by adding the optional gravity constraint.

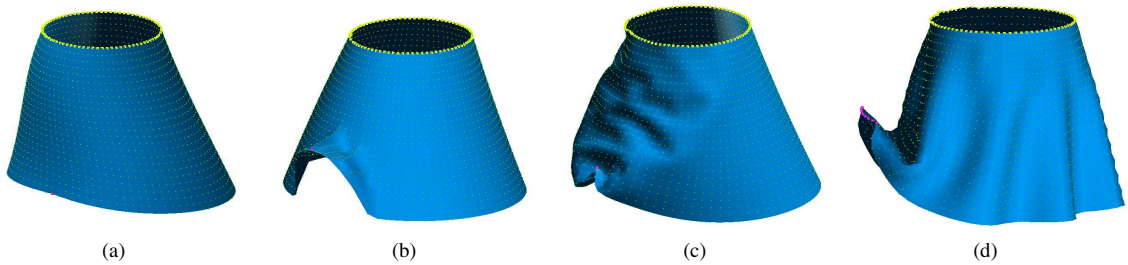

Figure 4. Compare our method to other approaches. The initial overskirt mesh with yellow fixing vertices and pink control vertices. (a) The membrane effect of Laplacian method [13]. (b) The result of space-based deformation method [17]. (c) The result of our method: the local geometric features (e.g., wrinkles and folds) are preserved. (d) The realistic effect with adding energy-based constraint.

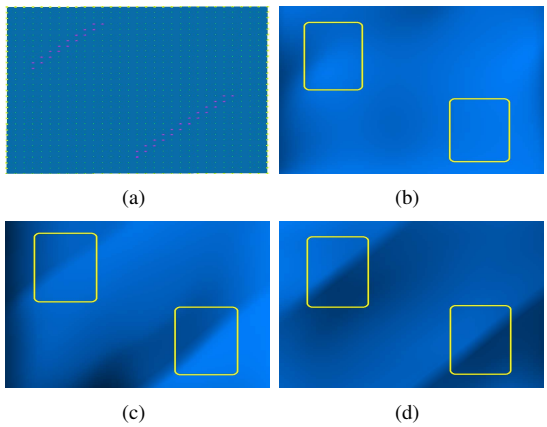

Figure 5. Draping cloth simulation: the initial piece of rectangular cloth (a) with pink control vertices and yellow fixed vertices around. The membrane effect (b). (c) and (d) exhibit the deformation effects from membrane to cloth when combining the length and angle constraints.

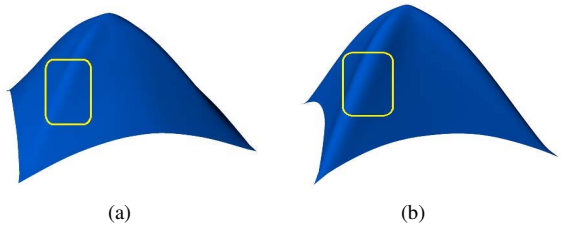

Figure 6. With the same parameters of $\alpha = 1, \beta = 1e-5, \delta = 1$, (a) exhibits the effect of the method [15], (b) shows the result of our algorithm.

With the different patterns of control vertices and parameters setting, the final and stable deformation shapes of cloths can be obtained. The simulation result comparing with the real photographs are shown in Fig. 8.

Tab. 1 lists the geometrical data of the input specific models in our experiments, the average initial time and the numerical solution time for our objective function. In addition, the iterations m is 10 for a suitable choice as well.

Table I
STATISTICS OF GEOMETRICAL DATA AND COMPUTING TIME.

Fig	Vertices	Triangles	Edges	Initial	Linear	Solution
1,3	961	1800	5520	0.156 s	0.0032 s	0.016 s
5,7	1326	2500	7650	0.373 s	0.0125 s	0.045 s
6,8	2601	2500	10200	1.154 s	0.0485 s	0.245 s
4	2000	3840	11680	0.863 s	0.0415 s	0.191 s

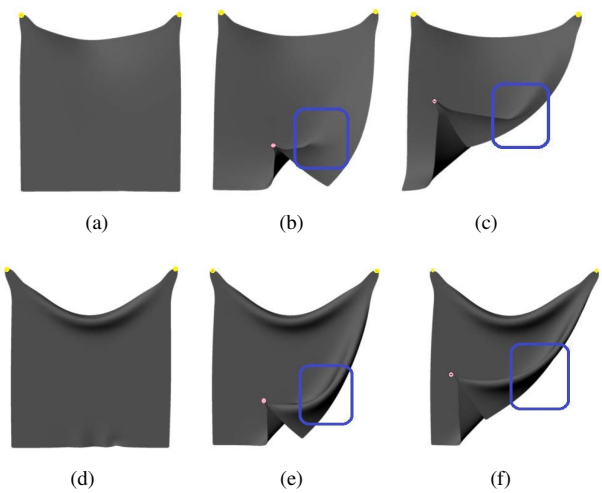

Figure 7. (a) The initial cloth without gravity energy with the fixing vertices(yellow color). (d) The initial cloth with gravity energy. During suspension, two kinds of different fabric drape details can be obviously shown. By adjusting the parameters, the silk effects are shown in (d)-(f), note that the different wrinkles and folds marked with blue rectangular. ($\alpha = 1, \beta = 1e-3, \delta = 1$). The hard material effects are performed as shown in (b)-(c). ($\alpha = 1, \beta = 5e-6, \delta = 1$) .

V. CONCLUSIONS AND FUTURE WORK

This paper has presented a novel method based on differential domain constraints to simulating cloth. We proposed nonlinear deformation energy which can obtain the local vertices transformation of the target cloth meshes automatically in the iterative procedure of energy optimization. In order to further improve the geometric deformable control, we also aimed at using edge and offset based differential quantities to achieve the local shape and the rich cloth effects. Our method combines the efficiency and control of geometric methods together with material effects. It can generate physically plausible results and distinguish different cloth materials with many interesting wrinkling and folding effects.

Our method has several unconsidered key problems. It is only able to find the final shape to model similar cloth behavior, but not its dynamic behavior. And it cannot be suitable for cloth animation or varying vertices constrained through time to deal with collisions. In the future work,

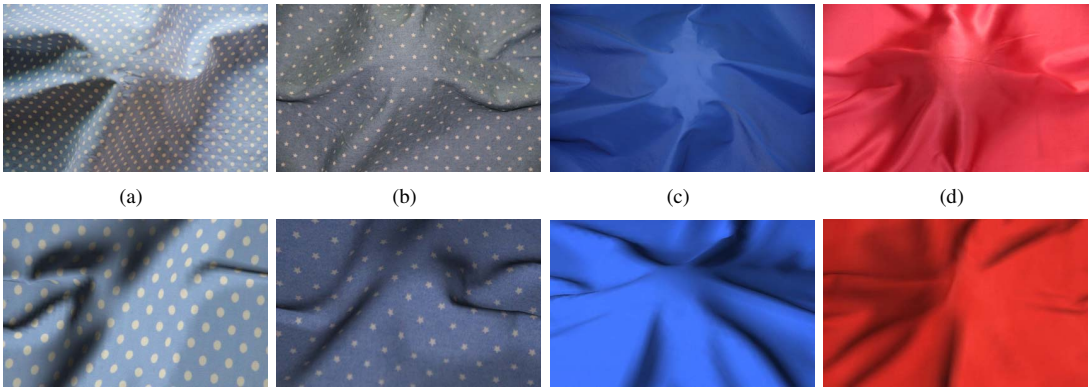

Figure 8. Simulation results of four types of cloth materials. The real photographs of four materials are shown in (a)-(d) and the simulation results of canvas, denim, nylon and silk with texture mapping shown in (e)-(h) respectively.

we would like to improve our cloth modeling for more cloth materials and solve existing limitations in our current method as described above.

VI. Acknowledgements

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